Experiment 16: Write the Python Program to Implement Feed Forward Neural Network

Aim: To implement a Feed Forward Neural Network using Python to predict output based on input values.

Procedure:

1. Import the required library.
2. Create a sample dataset.
3. Separate input features and target outputs.
4. Create a neural network model.
5. Train the model using the dataset.
6. Predict and display the output.

Input:

```
from sklearn.neural_network import MLPClassifier

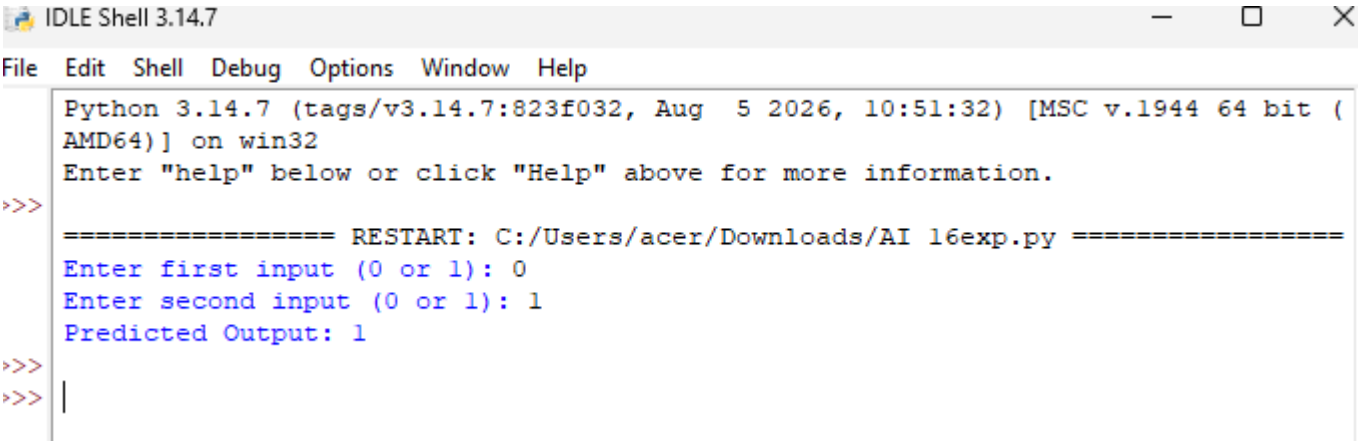
X = [[0, 0], [0, 1], [1, 0], [1, 1]]
y = [0, 1, 1, 0]

model = MLPClassifier(hidden_layer_sizes=(4,), solver='lbfgs', max_iter=10000, random_state=1)
model.fit(X, y)

a = int(input("Enter first input (0 or 1): "))
b = int(input("Enter second input (0 or 1): "))

print("Predicted Output:", model.predict([[a, b]])[0])
```

Output:



Result: Thus, a Feed Forward Neural Network was implemented successfully using Python.